

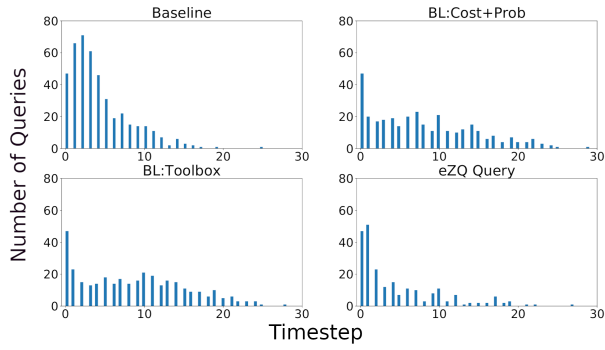


Figure 4: Histogram of number of queries (y-axis) executed per timestep (x-axis) by each method with the Boltzmann distribution of the worker’s distances to goals and a per-station cost of 0.

work and the heuristics described above. We demonstrate that *eZQ Query* is able to effectively leverage the additional information from our generalizations to obtain a better performance over previous work and the suggested heuristics, in terms of marginal cost (Definition 7). Figure 3 shows the marginal cost averaged over 100 domain instances with different per-station costs (x-axis) when the probability distribution of the goals of the worker is the Boltzmann distribution over the worker’s distances to each goal. This distribution means that the worker is more likely to choose goals that are farther from it. *eZQ Query* performs similarly to the heuristics when the per-station cost is 0, but dramatically outperforms all other methods as this value increases. Additional results showing *eZQ Query* under additional goal distributions can be seen in the Appendix.

We also provide an analysis of how the *eZQ Query* method is achieving better performance than other methods. Figure 4 shows a histogram of the number of queries executed per timestep by each method. As shown in the histogram, *eZQ Query* tends to ask more queries in the earlier timesteps compared to *BL:Toolbox* and *BL:Cost+Prob*. This increase is because *eZQ Query* is focused on learning the worker’s true goal with minimal cost and is better able to leverage information about the probability distribution over goals to make informed queries, and therefore finds the correct goal more quickly than other methods, but also takes longer before it knows an optimal action. In addition, we found that as the per-station cost increases from 0 to 0.5, the total number of queries executed by *eZQ Query* over all simulations decreases by 23%, showing that *eZQ Query* executes fewer queries when the cost is higher. Finally, there is an increased computational cost of using *eZQ Query* over other approaches and the proposed heuristics. While calculating EDP is expensive and takes several orders of magnitude longer than the rest of the querying algorithm (several hours per domain on average), these values can be computed a priori regardless of the teammate’s actions. As such, the following results are under the assumption that the EDP computation is performed in advance, and the following time measurements do not include these offline compu-

tations. On average, all heuristic methods took  $< 0.23$  seconds to complete each simulation, while *eZQ Query* took on average 8.9 seconds on an Ubuntu 16.04 LTS Intel core i7 2.5 GHz, with the genetic algorithm taking on average 6.1 seconds to run. In practice, the increased time should not be a major detriment. If a robot is communicating with either a human or another robot, the major bottleneck is likely to be the communication channel (e.g. speech, network speed, decision making of the other agent) rather than this time. In addition, when using a genetic algorithm for optimization as we do in this paper, the *eZQ Query* computation should only grow in terms of  $O(|G|^2 \log(|G|))$  with the number of goals (assuming that EDP is precomputed ahead of time and that the number of members and generations do not grow with the number of goals).

## Discussion and Future Work

In this paper, we investigated a new metric to quantify ambiguity of teammate policies in ad hoc teamwork settings, by estimating the expected divergence point between different policies a teammate might possess. We then utilized this metric to construct a new ad hoc agent that reasons both about *when* it is beneficial to query, but also about *what* is beneficial to query about in order to reduce the ambiguity about its teammate’s goal. Our empirical results show that regardless of the goal-choosing policy of the worker and a varying query cost model, *eZQ Query* remains more effective than any of the other methods tested, and even when querying is almost never beneficial, it is still able to adapt and obtain performance that is consistently better than *Never Query*. The scope of this work is limited to SOMALI CAT problems. In addition, our current methods are designed to work in relatively simple environments with finite state spaces and a limited number of goals. However, the EDP formalization opens up new avenues for investigating other complicated SOMALI CAT scenarios and other CAT scenarios, such as those in which an agent can advise or share its beliefs with its teammates. We conjecture that the *eZQ* algorithm can be modified relatively easily to address such challenges, as long as the ego agent remains the initiator of the communication. For instance, EDP may be able to be calculated in domains with larger and continuous state spaces by leveraging more sophisticated RL techniques than the policy evaluation algorithm. It might be more challenging to extend this work to domains in which the teammate is the one to initiate the communication, as other works have investigated in the context of reinforcement learning agents (?). Nonetheless, this work provides the means to investigate collaborations in ad hoc settings in new contexts, while presenting concrete solutions for planning in SOMALI CAT settings.

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